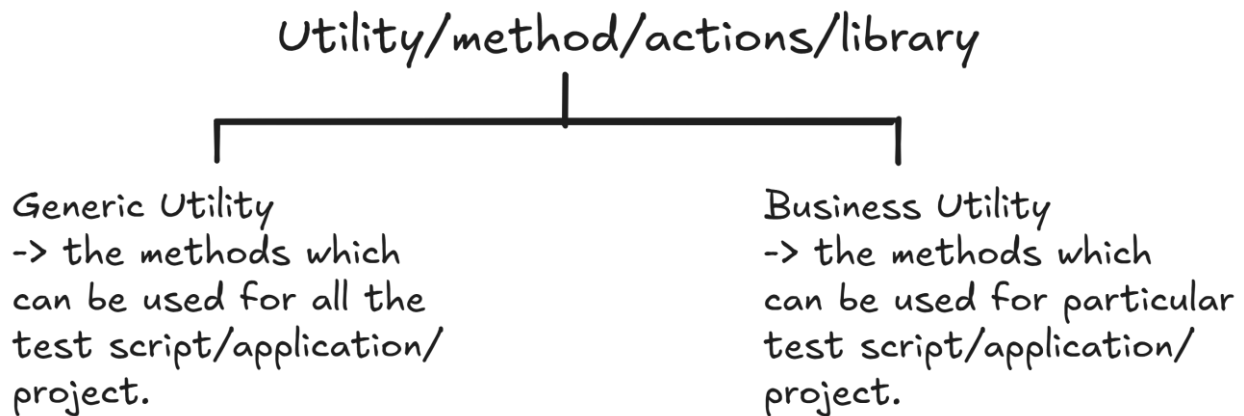


Generic Utility



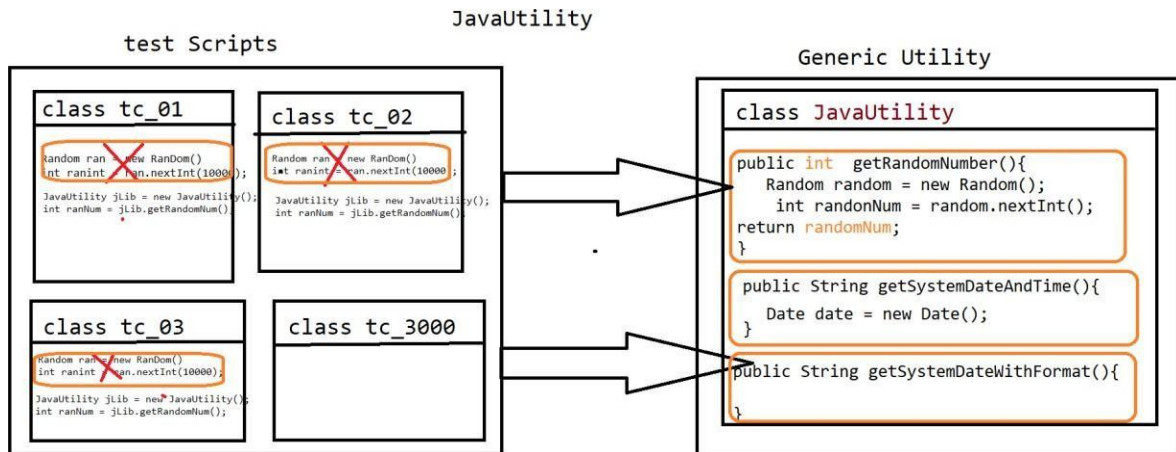
What is Generic components in Automation Framework?

- it's one of the automation framework components which is common for all the application
- its collection of generic class contains reusable methods / libraries
- The methods which can be used to any application is called Generic/common methods

What is the advantages of Generic components?

1. Reusability of code
2. Code Optimization
3. Test script development is faster
4. Test scripts Code readability
5. Generic libraries are common to all automation projects
6. Avoid duplicate Code
7. no need to remember the syntax every time, just create once & use multiple times

1. Java Utility Libraries

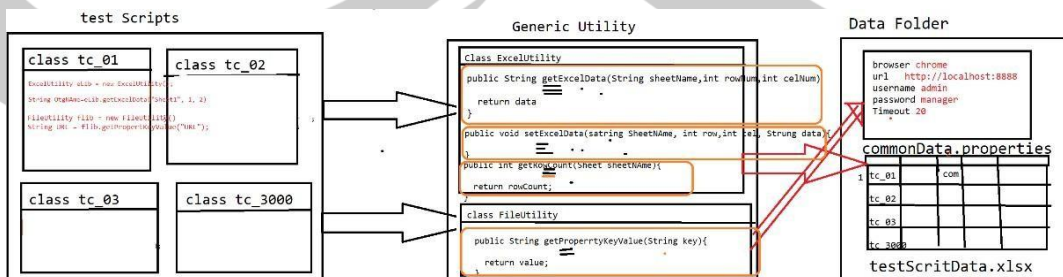


➔ Java Utility is one class in generic component, which contain java specific methods which can be used across the test Scripts / Application

➔ its contains several generic reusable methods like

- ⇒ `getRandomNum()` : it's used to generate random number for every invocation
- ⇒ `getSystemDate()` : it's used to generate system date and time

2. File Utility



- ⇒ As per the rule of automation, data should not be hardcoded with in the test scripts, so that to get the data from external file like Excel & .propertes file We go for ExcelUtility & FileUtility
- ⇒ Excel Utility class is developed using apache Poi libraries, which is used to read the data from Excel
- ⇒ FileUtility is used to get the data from .properties file

Version v1.0 - Raw file with hardcoded data

```
package crm.organization;

import java.io.IOException;
import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.interactions.Actions;

public class CreateOrgTest {
    public static void main(String[] args) throws InterruptedException,
    IOException {

//        open browser
        WebDriver driver = null;

        String browser = "chrome";

        if (browser.equals("chrome")) {
            driver = new ChromeDriver();
        } else if (browser.equals("edge")) {
            driver = new EdgeDriver();
        } else if (browser.equals("firefox")) {
            driver = new FirefoxDriver();
        } else {
            driver = new ChromeDriver();
        }

        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(15));

//        login
        driver.get("https://localhost:8888");

        WebElement un = driver.findElement(By.name("user_name"));
        un.sendKeys("admin");

        WebElement pwd = driver.findElement(By.name("user_password"));
        pwd.sendKeys("password");

        WebElement loginButton = driver.findElement(By.id("submitButton"));
        loginButton.click();

//        create organization
        driver.findElement(By.LinkText("Organizations")).click();
    }
}
```

```

        driver.findElement(By.cssSelector("img[title='Create
Organization...']")).click();

//        fill the form
        String orgName = "qsp" + System.currentTimeMillis()/1000;
        WebElement orgField = driver.findElement(By.name("accountname"));

        orgField.sendKeys(orgName);

//        save
        driver.findElement(By.className("save")).click();

//        verification
        String actOrgName = driver.findElement(By.id("dtlview_Organization
Name")).getText();

        if (actOrgName.equals(orgName)) {
            System.out.println("org created successfullyyyy !!!");
        } else {
            System.out.println("Could not create organization");
        }

//        logout
        WebElement profile =
driver.findElement(By.cssSelector("img[src='themes/softed/images/user.PNG']"));
;

//        hover on profile
        Actions act = new Actions(driver);
        act.moveToElement(profile).build().perform();

//        click on sign out link
        driver.findElement(By.LinkText("Sign Out")).click();

//        close the browser
        Thread.sleep(3000);
        driver.quit();
    }
}

```

Version v2.0 - DDT

```
package crm.organization;

import java.io.FileInputStream;
import java.io.FileReader;
import java.io.IOException;
import java.time.Duration;

import org.apache.poi.ss.usermodel.Cell;
import org.apache.poi.ss.usermodel.Row;
import org.apache.poi.ss.usermodel.Sheet;
import org.apache.poi.ss.usermodel.Workbook;
import org.apache.poi.ss.usermodel.WorkbookFactory;
import org.json.simple.JSONObject;
import org.json.simple.parser.JSONParser;
import org.json.simple.parser.ParseException;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.interactions.Actions;

public class CreateOrgTest {
    public static void main(String[] args) throws InterruptedException,
    IOException, ParseException {

//        get the data from json file
        FileReader fr = new FileReader("./src/test/resources/commondata.json");
        JSONParser parser = new JSONParser();
        Object obj = parser.parse(fr);
        JSONObject jsonObj = (JSONObject) obj;
        String browser = jsonObj.get("bro").toString();
        String url = jsonObj.get("url").toString();
        String username = jsonObj.get("un").toString();
        String password = jsonObj.get("pwd").toString();

//        get the data from Excel file
        FileInputStream fis = new
FileInputStream("./src/test/resources/demo.xlsx");
        Workbook wb = WorkbookFactory.create(fis);
        Sheet sh = wb.getSheet("org");
        Row row = sh.getRow(2);
        Cell cell = row.getCell(0);
        String orgName = cell.getStringCellValue() +
System.currentTimeMillis()/1000;

//        open browser
        WebDriver driver = null;

        if (browser.equals("chrome")) {
            driver = new ChromeDriver();
```

```

    } else if (browser.equals("edge")) {
        driver = new EdgeDriver();
    } else if (browser.equals("firefox")) {
        driver = new FirefoxDriver();
    } else {
        driver = new ChromeDriver();
    }

    driver.manage().window().maximize();
    driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(15));

//    login
    driver.get(url);

    WebElement un = driver.findElement(By.name("user_name"));
    WebElement pwd = driver.findElement(By.name("user_password"));
    WebElement loginButton = driver.findElement(By.id("submitButton"));

    un.sendKeys(username);
    pwd.sendKeys(password);
    loginButton.click();

//    create organization
    driver.findElement(By.LinkText("Organizations")).click();
    driver.findElement(By.cssSelector("img[title='Create
Organization...']")).click();

//    fill the form
    WebElement orgField = driver.findElement(By.name("accountname"));

    orgField.sendKeys(orgName);

//    save
    driver.findElement(By.className("save")).click();

//    verification
    String actOrgName = driver.findElement(By.id("dtlview_Organization
Name")).getText();

    if (actOrgName.equals(orgName)) {
        System.out.println("org created successfullyyyy !!!");
    } else {
        System.out.println("Could not create organization");
    }

//    logout
    WebElement profile =
driver.findElement(By.cssSelector("img[src='themes/softed/images/user.PNG']"));

//    hover on profile
    Actions act = new Actions(driver);
    act.moveToElement(profile).build().perform();

//    click on sign out link
    driver.findElement(By.LinkText("Sign Out")).click();

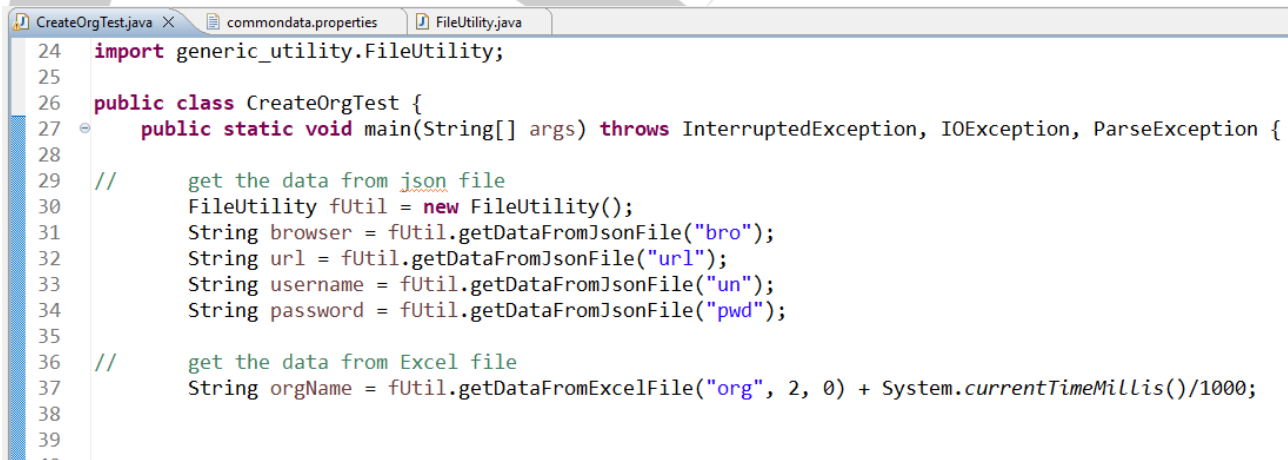
```

```
//      close the browser
      Thread.sleep(3000);
      driver.quit();
    }
}
```

Version v3.0 - Optimized Test scripts



```
23
24 public class CreateOrgTest {
25     public static void main(String[] args) throws InterruptedException, IOException, ParseException {
26
27         //      get the data from json file
28         FileReader fr = new FileReader("./src/test/resources/commondata.json");
29         JSONParser parser = new JSONParser();
30         Object obj = parser.parse(fr);
31         JSONObject jsonObj = (JSONObject) obj;
32         String browser = jsonObj.get("bro").toString();
33         String url = jsonObj.get("url").toString();
34         String username = jsonObj.get("un").toString();
35         String password = jsonObj.get("pwd").toString();
36
37         //      get the data from Excel file
38         FileInputStream fis = new FileInputStream("./src/test/resources/demo.xlsx");
39         Workbook wb = WorkbookFactory.create(fis);
40         Sheet sh = wb.getSheet("org");
41         Row row = sh.getRow(2);
42         Cell cell = row.getCell(0);
43         String orgName = cell.getStringCellValue() + System.currentTimeMillis()/1000;
44
45
46
```



```
24 import generic_utility.FileUtility;
25
26 public class CreateOrgTest {
27     public static void main(String[] args) throws InterruptedException, IOException, ParseException {
28
29         //      get the data from json file
30         FileUtility fUtil = new FileUtility();
31         String browser = fUtil.getDataFromJsonFile("bro");
32         String url = fUtil.getDataFromJsonFile("url");
33         String username = fUtil.getDataFromJsonFile("un");
34         String password = fUtil.getDataFromJsonFile("pwd");
35
36         //      get the data from Excel file
37         String orgName = fUtil.getDataFromExcelFile("org", 2, 0) + System.currentTimeMillis()/1000;
38
39
40
```

```
package crm.organization;
```

```
import java.io.FileInputStream;
import java.io.FileReader;
import java.io.IOException;
import java.time.Duration;

import org.apache.poi.ss.usermodel.Cell;
import org.apache.poi.ss.usermodel.Row;
import org.apache.poi.ss.usermodel.Sheet;
import org.apache.poi.ss.usermodel.Workbook;
import org.apache.poi.ss.usermodel.WorkbookFactory;
import org.json.simple.JSONObject;
import org.json.simple.parser.JSONParser;
import org.json.simple.parser.ParseException;
```

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.interactions.Actions;

import generic_utility.FileUtility;

public class CreateOrgTest {
    public static void main(String[] args) throws InterruptedException,
IOException, ParseException {

//        get the data from json file
        FileUtility fUtil = new FileUtility();
        String browser = fUtil.getDataFromJsonFile("bro");
        String url = fUtil.getDataFromJsonFile("url");
        String username = fUtil.getDataFromJsonFile("un");
        String password = fUtil.getDataFromJsonFile("pwd");

//        get the data from Excel file
        String orgName = fUtil.getDataFromExcelFile("org", 2, 0) +
System.currentTimeMillis() / 1000;

//        open browser
        WebDriver driver = null;

        if (browser.equals("chrome")) {
            driver = new ChromeDriver();
        } else if (browser.equals("edge")) {
            driver = new EdgeDriver();
        } else if (browser.equals("firefox")) {
            driver = new FirefoxDriver();
        } else {
            driver = new ChromeDriver();
        }

        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(15));

//        login
        driver.get(url);

        WebElement un = driver.findElement(By.name("user_name"));
        WebElement pwd = driver.findElement(By.name("user_password"));
        WebElement loginButton = driver.findElement(By.id("submitButton"));

        un.sendKeys(username);
        pwd.sendKeys(password);
        loginButton.click();

//        create organization
        driver.findElement(By.LinkText("Organizations")).click();
        driver.findElement(By.cssSelector("img[title='Create
Organization...']")).click();

```

```

//      fill the form
//      String orgName = "qsp" + System.currentTimeMillis()/1000;
//      WebElement orgField = driver.findElement(By.name("accountname"));

//      orgField.sendKeys(orgName);

//      save
//      driver.findElement(By.className("save")).click();

//      verification
//      String actOrgName = driver.findElement(By.id("dtlview_Organization
//      Name")).getText();

//      if (actOrgName.equals(orgName)) {
//          System.out.println("org created successfullyyyy !!!");
//      } else {
//          System.out.println("Could not create organization");
//      }

//      logout
//      WebElement profile =
//      driver.findElement(By.cssSelector("img[src='themes/softed/images/user.PNG']"));

//      hover on profile
//      Actions act = new Actions(driver);
//      act.moveToElement(profile).build().perform();

//      click on sign out link
//      driver.findElement(By.LinkText("Sign Out")).click();

//      close the browser
//      Thread.sleep(3000);
//      driver.quit();
}
}

```

3. WebDriver Utility Libraries

